

Real-Time Web QA Validator

Automated Cross-Checking master specifications against live Wix web pages

Executive Summary: The Real-Time Web QA Validator is a fully dynamic QA checker tool designed to crawl and validate live websites (such as Wix, WordPress, and custom pages) against client-approved copy documents (PDFs, Word documents, text/markdown). It parses the document's sections and pages automatically and verifies that titles, headers, body text, buttons, forms, and SEO tags are correctly implemented on the website.

1. Technology Stack

The application is built using a modern full-stack JavaScript environment, designed to run both locally and on cloud environments:

Layer	Technologies Used	Purpose
Frontend	React 19, TypeScript, Vite, CSS Glassmorphism	Builds a high-fidelity dark-themed, responsive dashboard.
Backend	Node.js, Express, TSX	Handles PDF/Docx uploads and manages validation requests.
Document Parsing	pdf-parse, mammoth.js	Extracts text hierarchy from Word files and PDF streams.
Scraping & Crawling	Playwright, Cheerio, Axios	Crawls site links and extracts DOM content.
Deployment	Vercel, Render	Vercel hosts the React UI, Render hosts the Express API.

2. Key Capabilities & Implemented Features

- Fully Dynamic Checks:** The tool parses and validates *any* document structure against *any* website. If page headers or markers are missing, it dynamically groups sections under default page buckets.
- Wix Button & Anchor Validation:** Wix anchors (e.g. #action or #contact) and lead forms are scanned. If they are mapped to standard links in the DOM, the engine falls back to check page

links to prevent false warnings.

- **Spelling & Discrepancy Audits:** Employs bigram text similarity algorithms to detect small typos and copy differences, outputting clear correction guidelines.
- **SEO & Metadata Checks:** Validates Page Title, Meta Description, H1 titles, and Image Alt text.
- **Interactive UI Elements:** Features a high-fidelity glassmorphic card design, responsive columns, a real-time glowing loading bar with a 3-step scanner progress indicator, and custom styling filters.

3. Performance Optimizations

Low-Resource Fail-Safe (512MB RAM Bypass): Running Chromium browsers with Playwright consumes more than 1GB of memory. Render's free tier has a strict 512MB limit, causing server crashes. The engine automatically detects production environments and switches from Playwright to a fast Cheerio static crawler, ensuring validations finish in under 5 seconds with zero server hangs!

4. Achievements & Completed Work

Module	Enhancement Done	Status
HTML & Text Parsing	Filtered split PDF lines like (Hero image) and page numbers (e.g. - 3 of 8 --).	COMPLETED
Button Verifier	Added Wix anchor support and links-fallback to avoid Not Found errors.	COMPLETED
Responsive Design	Made dashboard grids, navbar, and footer stack beautifully on all mobile screens.	COMPLETED
Layout UI	Swapped expected spec copy to the left and live website copy to the right.	COMPLETED
Progress Loader	Added glowing neon loading bar with active step status indicators.	COMPLETED